EMG Signal Interpretation Using LLMs

Complete Workflow for Clinical EMG Analysis

Signal Quality · Signal Drift · Fatigue · Muscle Activation

Feature Extraction Pipeline · LLM Prompts · RAG Rules · Reference Sources

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1. Executive Summary

This document provides a complete, production-ready workflow for interpreting surface electromyography (sEMG) signals using Large Language Models (LLMs). The system is designed for clinical use, where doctors receive structured, actionable insights generated by an LLM that has been provided with extracted EMG features and domain-specific rules.

The pipeline processes raw EMG data from 8 Delsys Avanti sensors recorded across 44 patients, each performing 5 gestures for 20 minutes. Raw signals are segmented using a multi-scale windowing strategy: long windows (20 seconds) for drift and fatigue trending, and short windows (100–250 ms) for activation onset/offset detection. A comprehensive feature set of 30+ features per sensor is extracted covering time-domain, frequency-domain, cross-channel, and nonlinear measures. These features are structured into a compact, token-efficient JSON format that is injected into the LLM prompt alongside clinical interpretation rules retrieved via RAG.

Four clinical assessment workflows are covered: (1) Signal Quality Assessment, (2) Signal Drift Detection, (3) Fatigue Assessment, and (4) Muscle Activation Analysis. Each workflow includes specific features, interpretation rules, ready-to-use prompt templates, and recommended RAG knowledge base entries. The workflows are sequential: signal quality must pass before drift analysis, and both must be acceptable before fatigue or activation inference.

> **Safety Note: The LLM must not be used as a standalone diagnostic tool. It is intended for summarization, explanation, and highlighting potential issues for clinician review. Always preserve raw data and computed features for audit. Implement human-in-the-loop review before reports reach the treating physician.**

The recommended LLM models include GPT-4o-mini (best cost/quality ratio), Qwen 2.5 7B/14B (open-source local deployment), Claude Haiku 4.5 (API, fast), or Llama 3.1 8B Instruct. The key requirement is strong JSON output fidelity and adherence to rule-based reasoning.

2. System Architecture Overview

2.1 End-to-End Pipeline

The system operates in five stages:

Stage 1 — Data Ingestion: Raw CSV files from Delsys Trigno Avanti sensors (8 channels, ACC/GYRO/EMG per sensor) are loaded. Only the EMG millivolt and corresponding time columns are retained. Timestamps are synchronized across channels.

Stage 2 — Preprocessing: DC offset removal (subtract segment mean), band-pass filtering (20–450 Hz, Butterworth 4th order), optional notch filter at 50/60 Hz if powerline interference detected, and rectification + low-pass smoothing (for activation envelope features only). Normalization to MVC or internal reference period when available. Both raw and filtered signals are stored for auditability.

Stage 3 — Feature Extraction: Multi-scale windowing is applied. Long windows (20 s, non-overlapping) produce 60 segments for drift/fatigue trending. Short windows (250 ms) produce ~4800 windows for activation onset/offset detection. For each window and sensor, 30+ features are computed spanning time-domain, frequency-domain, signal quality, drift, and cross-channel metrics.

Stage 4 — LLM Prompt Construction: Features are aggregated into a compact session-summary JSON with short keys to minimize tokens. RAG rules matching the analysis task are retrieved and prepended as system context. Deterministic rule-based flags are pre-computed before the LLM call.

Stage 5 — LLM Inference & Report: The LLM generates a strict JSON clinical interpretation. Output is logged with model version and prompt hash for traceability. The report is formatted for physician review.

2.2 Multi-Scale Windowing Strategy

Different EMG phenomena operate at different time scales, so a single window size is suboptimal:

Short windows (100–250 ms): Used for activation onset/offset detection, burst identification, and transient artifact detection. These capture sub-second phenomena that 20 s windows would obscure.

Long windows (20 s): Used for drift slope estimation, fatigue trending (MDF over time), and steady-state feature computation. These are robust to momentary fluctuations.

The feature extraction script produces two output files: features_long.csv (per 20 s segment) and features_short.csv (per 250 ms window for activation analysis).

2.3 Data Dimensions

Patients: 44 | Gestures per patient: 5 | Session duration: 20 min (1200 s) | Sensors: 8 (Delsys Avanti) | Sampling rate: ~1259 Hz | Long segment: 20 s (60 per session) | Short window: 250 ms (~4800 per session) | Features per sensor per long segment: 30+ | Total features per long segment: 240+.

2.4 Preprocessing Recommendations

For reproducibility, encode preprocessing explicitly even if working from pre-filtered files:

(1) Synchronize timestamps across channels; interpolate missing timestamps conservatively.

(2) Remove DC offset per window (subtract mean) before spectral features.

(3) Band-pass filter: 20–450 Hz (Butterworth 4th order). Some protocols use 10–400 Hz.

(4) Notch filter at 50/60 Hz (+ harmonics) only when PowerlineRatio > 0.10.

(5) Rectify + low-pass smooth only for activation envelope. Use 10–25 ms time constant when timing matters, up to 250 ms for amplitude trends.

(6) Normalize to MVC if available; otherwise use internal reference (first stable minute). Report as %MVC or %baseline.

Store both raw and filtered signals with filter parameters (type, cutoffs, order) for auditability.

2.5 Recommended LLM Models

For structured feature interpretation, smaller models suffice. Options: GPT-4o-mini (best cost/quality), Qwen 2.5 7B/14B (open-source local), Claude Haiku 4.5 (API, fast), Llama 3.1 8B Instruct. Key requirement: strong JSON output fidelity and rule-based reasoning adherence. All outputs must be strict JSON (no markdown).

3. Feature Extraction Pipeline

3.1 Design Principles

A small LLM works best when given ~50–300 numbers plus text labels, not thousands of raw samples. Features are task-driven: per-sensor aggregates, slopes/percent change over time, and a short list of notable anomaly segments.

3.2 Complete Feature List

Time-Domain Features (11 features per sensor per window)

RMS (Root Mean Square) — Overall signal amplitude in mV. Primary measure of contraction intensity.

MAV (Mean Absolute Value) — Average rectified amplitude. Robust to outliers. Correlates with force.

IEMG (Integrated EMG) — Sum of absolute values. Total muscular effort in the window.

Waveform Length — Cumulative absolute difference between consecutive samples. Signal complexity.

Zero Crossings (ZC) — Signal crossings of the mean (with noise threshold). Motor unit firing rate proxy.

Slope Sign Changes (SSC) — Number of slope direction changes (with noise threshold). Complementary to ZC for capturing signal complexity.

Willison Amplitude (WAMP) — Count of consecutive differences exceeding 0.01 mV. Motor unit recruitment sensitivity.

Peak Amplitude — Maximum absolute value. Burst/artifact/clipping detection.

Variance — Signal power and recruitment variability.

Skewness — Amplitude distribution asymmetry. Non-zero values may indicate spikes, clipping, or non-physiological components.

Kurtosis — Tail heaviness of amplitude distribution. High kurtosis (>10) indicates spike artifacts. Low kurtosis may indicate clipping.

Frequency-Domain Features (12 features per sensor per window)

Mean Frequency (MNF) — Weighted average of power spectrum. Gold standard fatigue indicator with MDF.

Median Frequency (MDF) — Divides spectrum into equal power halves. More robust to noise than MNF.

Peak Frequency — Frequency with maximum power.

Total Power — Sum of power spectrum (20–450 Hz). Overall signal energy.

Low Band Power (15–45 Hz) — Increases during fatigue.

Mid Band Power (45–95 Hz) — Middle frequency energy.

High Band Power (95–250 Hz) — Decreases during fatigue.

Low-to-High Ratio (L/H) — Low/high band power ratio. Key composite fatigue index.

Spectral Entropy — Spectrum uniformity. Low values = single-frequency dominance.

Powerline Ratio — Power at 50/60 Hz ±1 Hz / total power. >0.10 = significant contamination.

Low-Frequency Artifact Ratio — Power(1–10 Hz)/Power(20–450 Hz). >0.30 = motion/cable artifact.

High-Frequency Noise Ratio — Power(>450 Hz)/Power(20–450 Hz). Elevated = HF noise.

Nonlinear Feature (1 feature)

Sample Entropy — Signal complexity/regularity. Decreases during fatigue (motor unit synchronization). Range 0.5–2.5.

Signal Quality Features (4 features)

SNR (dB) — Signal RMS / baseline noise RMS. <10 dB = poor.

Baseline Noise RMS — RMS of quietest 50 ms window. <5 μV for quality systems.

Clipping Ratio — Fraction near max amplitude. >0.01 = saturation.

Dropout/Flatline Percent — Constant-value or NaN runs. >0.5% or flatline >200 ms = possible disconnection.

Signal Drift Features (2 features)

Drift Slope — Linear regression slope of raw EMG per segment.

DC Offset — Segment mean. Drift across segments = electrode degradation.

Cross-Channel Features (4 features, across 8 sensors)

Activation Map — Each sensor's fraction of total RMS. Spatial activation distribution.

Concentration Index (Herfindahl) — Sum of squared activation fractions. High = focal; low = diffuse.

Inter-Channel Correlation — Mean off-diagonal Pearson correlation. >0.8 = crosstalk/common-mode noise.

Pattern Stability (Cosine Similarity) — Activation map similarity between consecutive windows. <0.7 = spatial reconfiguration.

3.3 Feature Selection Rationale

Features selected based on: EMGBench (Yang et al. 2024) for preprocessing benchmarks; SENIAM/ISEK standards for quality criteria; Phinyomark et al. (2012) for discriminative power; De Luca (1997, 2006) for signal decomposition; Chowdhury et al. (2013) for artifact characterization; Hodges & Bui (1996) for onset detection; and Vigotsky et al. (2018) for amplitude interpretation limitations.

4. Workflow 1: Signal Quality Assessment

4.1 Purpose

Determine whether the recorded EMG is of sufficient quality for clinical interpretation. This is the gating step — all subsequent analyses depend on quality being adequate. Artifact types to detect: powerline interference, motion artifact, crosstalk, dropouts, clipping, poor electrode contact.

4.2 Key Features

Primary: SNR_dB, BaselineNoiseRMS, PowerlineRatio, ClippingRatio, DropoutPct, spectral_entropy.

Supporting: LF_artifact_ratio, HF_noise_ratio, kurtosis, inter-channel correlation, DC offset.

Compute on short 1–2 s windows to detect intermittent artifacts, then aggregate per sensor.

4.3 Interpretation Rules (for RAG/Prompt)

Rule Q1 — Baseline Noise: Resting baseline noise RMS below 0.005 mV (5 µV) is good. Above 0.010 mV = POOR QUALITY (bad skin prep or electrode contact).

Rule Q2 — SNR Threshold: >15 dB reliable. 10–15 dB MARGINAL. <10 dB UNRELIABLE.

Rule Q3 — Powerline Contamination: PowerlineRatio >0.10 = significant PLI. Recommend notch/adaptive cancellation; check grounding.

Rule Q4 — Clipping/Saturation: ClippingRatio >0.01 or >0.5% samples at extreme percentiles = amplifier saturation. Reduce gain.

Rule Q5 — Spectral Shape: Broadband 10–500 Hz expected, peaking 50–150 Hz. Spectral entropy <3.0 bits = single-frequency artifact.

Rule Q6 — Cross-talk: Opposite-side sensors correlated >0.8 during unilateral gesture = crosstalk. Review electrode spacing.

Rule Q7 — Motion Artifact: DC offset shift >0.5 mV between segments + low-freq dominance (<5 Hz) = motion artifact. Exclude segments.

Rule Q8 — Dropout/Flatline: DropoutPct >0.5% or flatline >200 ms = possible disconnection. Mark segment unusable.

Rule Q9 — LF Artifact: LF_artifact_ratio (1–10 Hz / 20–450 Hz) >0.30 = motion/cable artifact. Check fixation; consider high-pass 20 Hz.

Rule Q10 — Anomalous Distribution: Kurtosis >10 = spike artifacts. Unusually high ZC/SSC relative to RMS = noise-dominated signal.

4.4 RAG Knowledge Sources

(1) SENIAM Guidelines — Electrode placement and skin preparation.

(2) ISEK Standards (Merletti) — Bandpass, noise, CMRR, definitions.

(3) Konrad (2005) 'The ABC of EMG' — Practical quality assessment.

(4) Tankisi et al. (2020) IFCN consensus on hardware/quality.

(5) Merletti & Parker (2004) Chapters 2–4 on signal quality.

(6) Campbell et al. (2022) IEEE review of quality analysis techniques.

(7) Chowdhury et al. (2013) Surface EMG signal processing — PLI, motion artifact, crosstalk, noise reduction.

5. Workflow 2: Signal Drift Detection

5.1 Purpose

Detect slow or sudden non-physiological changes over time: electrode-skin impedance evolution, perspiration, electrode displacement, gel drying, adherence changes, temperature effects. Must distinguish (a) contact/placement drift, (b) physiological fatigue, (c) intentional gesture variation.

5.2 Key Features

Primary: DriftSlope, DCOffset, RMS slope vs time, change-point indicators.

Supporting: BaselineNoiseRMS trend, SNR trend, MDF slope (fatigue separation), activation map cosine similarity, early-vs-late feature distributions.

5.3 Interpretation Rules (for RAG/Prompt)

Rule D1 — Drift Rate: DriftSlope >0.001 mV/s = baseline wander. Mild: 0.001–0.005. Moderate: 0.005–0.01. Severe: >0.01 mV/s.

Rule D2 — DC Offset Trend: Monotonic change >0.05 mV over 5+ consecutive segments = systematic drift.

Rule D3 — Noise Floor Drift: BaselineNoiseRMS increase >50% from first to last 5 segments = interface degradation.

Rule D4 — Impedance Drift: Simultaneous noise increase + SNR decrease = rising impedance. Common after 15–20 min.

Rule D5 — Sudden Shift: DCOffset change >0.1 mV between consecutive segments = electrode displacement. Flag for exclusion.

Rule D6 — Perspiration: Gradual noise decrease in first 5–10 min is normal (skin hydration). Concern if it then suddenly increases.

Rule D7 — Warm-Up Stabilization: Electrode-skin impedance changes during first 10–15 min (Sousa et al. 2023). Use minute 15+ as stable baseline for trend analysis.

Rule D8 — Amplitude-Frequency Dissociation: RMS change >20% but MDF change <5% = sweat/impedance drift, not fatigue. Fatigue requires concurrent MDF decline.

Rule D9 — Step Change Detection: CUSUM or change-point on RMS/activation map. Step >3×MAD = electrode displacement or cable tug. Annotate timing.

Rule D10 — Spatial Reconfiguration: Activation map cosine similarity <0.7 for 3+ segments = spatial pattern change. May indicate electrode shift or motor compensation.

5.4 RAG Knowledge Sources

(1) Sousa et al. (2023) Electrode-skin impedance variation over 20+ min sessions.

(2) Abdoli-Eramaki et al. (2012) Perspiration effects on sEMG amplitude (MF unchanged).

(3) Bhateja et al. (2019) Baseline correction methods.

(4) De Talhouet & Webster (1996) Skin-stretch motion artifacts.

(5) EMGBench (Yang et al. 2024) Distribution shifts from fatigue, perspiration, displacement.

6. Workflow 3: Fatigue Assessment

6.1 Purpose

Detect and quantify muscle fatigue over a 20-minute session. Fatigue manifests as: slowed conduction velocity shifting spectrum to lower frequencies, additional motor unit recruitment increasing amplitude, and progressive motor unit synchronization reducing signal complexity.

> **Prerequisite: Fatigue inference should ONLY run when signal quality is acceptable (no POOR sensors) and drift has been characterized/controlled. Uncorrected drift can mimic or mask fatigue signatures.**

6.2 Key Features

Primary: MDF, MNF, lh_ratio, RMS. Compute MDF slope in Hz/min (expected fatigue: -0.1 to -0.5 Hz/min, device-dependent).

Supporting: SampleEntropy, ZeroCross, Variance, spectral_entropy.

Spatial: activation map changes over time (fatigue causes compensatory spatial shifts).

6.3 Interpretation Rules (for RAG/Prompt)

Rule F1 — MDF Decline (Gold Standard): 10–20% decline from baseline = moderate fatigue. >20% = severe.

Rule F2 — MNF Parallel: MNF should decline with MDF. MNF-only decline may be noise/artifact.

Rule F3 — RMS Increase: Sustained contraction RMS increases 25–50% from motor unit recruitment. RMS up + MDF down = strong fatigue.

Rule F4 — L/H Ratio: Increase >30% from baseline = spectral compression consistent with fatigue.

Rule F5 — Entropy Decrease: SampleEntropy decline >15% = fatigue-related motor unit synchronization.

Rule F6 — ZC Decrease: >10% decrease supports fatigue (slowed conduction velocity).

Rule F7 — Onset Detection: Onset = 3+ consecutive MDF-declining segments. Report time in seconds.

Rule F8 — Fatigue Index: FI = (MDF_baseline - MDF_final) / MDF_baseline × 100%. Mild: 5–10%. Moderate: 10–20%. Severe: >20%.

Rule F9 — Multi-sensor Consistency: Multiple sensors fatiguing simultaneously strengthens finding.

Rule F10 — Confound Check: If SNR drops concurrently, electrode degradation is more likely than fatigue.

Rule F11 — MDF Slope Rate: Track Hz/min. Sustained -0.1 to -0.5 Hz/min = fatigue. Outside this range = possible artifact.

Rule F12 — Spatial Compensation: If concentration index decreases while total RMS is maintained, load is redistributing. Dominant sensor may change mid-session.

Rule F13 — Prerequisite Gate: Before concluding fatigue, confirm: (a) quality GOOD/MARGINAL, (b) drift None/Mild, (c) D8 rule excludes sweat-only amplitude change.

6.4 RAG Knowledge Sources

(1) De Luca (1997) Foundational fatigue-EMG reference. Electrode placement for spectral comparisons.

(2) Phinyomark et al. (2012) MNF/MDF fatigue review.

(3) Karthick et al. (2016) Time-frequency fatigue analysis.

(4) Al-Mulla et al. (2011) Non-invasive fatigue detection review.

(5) Merletti & Parker (2004) Chapters 7–9 on fatigue.

(6) EMGBench (Yang et al. 2024) Fatigue changes fiber activation and spatial distribution.

(7) ISEK Standards — MDF/MNF computation and normalization requirements.

(8) Gonzalez-Izal et al. (2012) Spectral indices during dynamic contractions.

7. Workflow 4: Muscle Activation Analysis

7.1 Purpose

Characterize pattern, intensity, timing, and consistency of activation across sensors. Detect onset/offset, quantify intensity, identify dominant muscles, and describe temporal profiles. Tracks rehabilitation progress.

7.2 Key Features

Long-window: RMS, MAV, IEMG, PeakAmp for activation level.

Short-window (250 ms): Rectified envelope, baseline mean/SD, onset/offset per sensor, duty cycle.

Cross-sensor: Activation map, top-K sensors, concentration index, pattern stability.

7.3 Interpretation Rules (for RAG/Prompt)

Rule A1 — Activation Level: RMS relative to session mean. Low: <30%. Moderate: 30–70%. High: 70–100%. Very High: >100%.

Rule A2 — Gesture Patterns: Flexion = S1–S4 (palmar). Extension = S5–S8. Deviations = placement error or compensation.

Rule A3 — Co-contraction: Agonist + antagonist RMS both >50% peak simultaneously. Normal for power grip; abnormal for fine motor.

Rule A4 — Consistency: CV of RMS. <20% consistent. 20–40% moderate. >40% difficulty maintaining gesture.

Rule A5 — Dominant Sensor: Highest average RMS. Report dominant-to-weakest ratio.

Rule A6 — Temporal Profile: Sustained, Ramp-up, Ramp-down, Intermittent, or Compensatory shift.

Rule A7 — Force Estimation: RMS has linear-to-quadratic relationship with force. Compare across patients.

Rule A8 — Motor Unit Recruitment: Higher WAMP/ZC = fast-twitch recruitment (increasing effort or fatigue).

Rule A9 — Onset Detection: Threshold = baseline mean + k×SD (k in [1,3]) with minimum duration 20–100 ms. Per Hodges & Bui (1996).

Rule A10 — Envelope Smoothing: Rectify + low-pass. Use ≤25 ms for timing accuracy; up to 250 ms for amplitude trends.

Rule A11 — False Onset Guard: Require consecutive windows above threshold. Single-window crossings = noise spikes.

Rule A12 — Duty Cycle: % time above activation threshold per sensor. Sustained vs intermittent classification.

Rule A13 — Normalization: Report as %MVC or %baseline. Without normalization, cross-session/subject comparisons are unreliable.

7.4 RAG Knowledge Sources

(1) Basmajian & De Luca (1985) 'Muscles Alive'.

(2) Konrad (2005) 'ABC of EMG'.

(3) Phinyomark et al. (2012) Feature selection for EMG.

(4) EMGBench (Yang et al. 2024) Gesture benchmarks.

(5) Hodges & Bui (1996) Onset detection: threshold (1–3 SD), duration (20–100 ms), filter effects.

(6) Vigotsky et al. (2018) Amplitude interpretation limits — what sEMG can/cannot conclude.

(7) ISEK Standards — Envelope smoothing, normalization.

(8) Nazmi et al. (2016) EMG classification review.

8. LLM Prompt Templates

Replace placeholders with actual data. All prompts require strict JSON output (no markdown).

8.1 System Prompt (Shared)

```
SYSTEM:
You are an expert clinical EMG analysis assistant. You interpret extracted
EMG signal features to provide structured clinical assessments for physicians.
Base analysis on provided features and rules only. Never fabricate data.
Do not invent new thresholds. If data insufficient, say so and lower confidence.
Do not diagnose disease. Role: summarization, explanation, flagging issues.
Return STRICT JSON matching the schema. No markdown.
```

8.2 Compact Input JSON Schema

Token-efficient session summary with short keys, aggregated metrics, slopes, and notable events:

```
{"meta":{"pid":"P1","gest":"wrist_flex","dur_s":1200,"fs_hz":1259},
 "win":{"long_s":20,"short_ms":250},
 "agg":{"S1":{"rms_mu":0.008,"rms_slope":-8e-7,"mdf_mu":105.2,
   "mdf_slope":-0.12,"pli_ratio":0.03,"lf_ratio":0.08,
   "clip_pct":0.0,"drop_pct":0.0},"...":"..."},
 "global":{"quality":{"grade":"B","issues":["mild_pli_S3"]},
   "drift":{"grade":"mild","type":["amplitude_drift"],"warmup_min":15},
   "fatigue":{"grade":"possible","evidence":["mdf_down_S2_S4"]},
   "activation":{"pattern":"stable","top":["S4","S3"],"duty":0.92}},
 "notable":[{"seg":12,"t0":220,"event":"step_change","sensors":["S6"]}]}
```

8.3 Signal Quality Prompt

```
USER: <RAG_SNIPPETS:QUALITY_RULES> {Q1-Q10} </RAG_SNIPPETS>
Session JSON: {SESSION_JSON}
Task: Assess signal quality per sensor + overall.
Output: {"overall_grade":"A|B|C|D|F",
  "per_sensor":[{"sensor":"S1","grade":"...","issues":[...],"evidence":[...]}],
  "segments_to_exclude":[...],"recommended_actions":[...],"confidence":0-1}
```

8.4 Signal Drift Prompt

```
USER: <RAG_SNIPPETS:DRIFT_RULES> {D1-D10} </RAG_SNIPPETS>
Session JSON: {SESSION_JSON}
Task: Detect drift. Distinguish contact vs fatigue vs gesture variation.
Output: {"overall_grade":"none|mild|moderate|severe",
  "type":["amplitude_drift","baseline_drift","step_shift","spatial_reconfig"],
  "per_sensor":[{"sensor":"S1","drift_pct":...,"likely_causes":[...]}],
  "notable_events":[...],"warmup_stable_from_s":...,"confidence":0-1}
```

8.5 Fatigue Prompt

```
USER: <RAG_SNIPPETS:FATIGUE_RULES> {F1-F13} </RAG_SNIPPETS>
Quality grade: {QUALITY_GRADE} | Drift grade: {DRIFT_GRADE}
Session JSON: {SESSION_JSON}
Task: Assess fatigue. Only proceed if quality+drift acceptable.
Output: {"fatigue_likelihood":"none|possible|likely",
  "fatigue_index_pct":...,"onset_estimate_s":null|number,
  "per_sensor":[{"sensor":"S1","mdf_slope_hz_min":...,"grade":"..."}],
  "spatial_compensation":true|false,"confounds":[...],"confidence":0-1}
```

8.6 Activation Prompt

```
USER: <RAG_SNIPPETS:ACTIVATION_RULES> {A1-A13} </RAG_SNIPPETS>
Session JSON: {SESSION_JSON}
Task: Summarize activation (timing, intensity, spatial pattern).
Output: {"overall_activation":"low|moderate|high",
  "top_sensors":[...],"per_sensor":[{"sensor":"S1","onset_s":...,"duty_cycle":...}],
  "co_contraction":true|false,"pattern":"sustained|ramp_up|...",
  "pattern_notes":[...],"confidence":0-1}
```

9. RAG Knowledge Base Rules

9.1 Embedding Strategy

Each rule has a separate chunk with metadata: {category, rule_id, source}. Retrieve top 10–15 matching the task. Strategy: 3–8 chunks based on deterministic flags (e.g., PLI detected → retrieve Q3 + Chowdhury 2013; MDF declining → retrieve F1, F2, F7, F8 + De Luca 1997).

9.2 Complete Rule Set Summary

Quality Rules (Q1–Q10): 10 rules — Section 4.3. Covers noise, SNR, PLI, clipping, spectral shape, crosstalk, motion artifact, dropouts, LF artifact, anomalous distribution.

Drift Rules (D1–D10): 10 rules — Section 5.3. Covers drift rate, DC offset, noise floor, impedance, sudden shifts, perspiration, warm-up, amplitude-frequency dissociation, change-points, spatial reconfiguration.

Fatigue Rules (F1–F13): 13 rules — Section 6.3. Covers MDF/MNF, RMS, L/H ratio, entropy, ZC, onset, index, multi-sensor, confounds, slope rate, spatial compensation, prerequisite gate.

Activation Rules (A1–A13): 13 rules — Section 7.3. Covers levels, gesture patterns, co-contraction, consistency, dominance, temporal profiles, force, recruitment, onset detection, envelope, false onset, duty cycle, normalization.

Total: 46 task-specific rules + 7 physiological context rules = 53 rules.

9.3 Physiological Context Rules

P1 — EMG Range: Meaningful 10–500 Hz. <10 Hz = motion artifact. >500 Hz = noise.

P2 — Motor Units: 100–1000 fibers each. Henneman size principle. Fast-twitch = higher freq EMG.

P3 — Conduction Velocity: Normal 3–6 m/s. Fatigue reduces it, shifting spectrum lower.

P4 — Typical Amplitude: 0–6 mV p-p. Forearm moderate grip: 0.1–1.0 mV RMS. <0.01 mV = not over muscle.

P5 — Session Phases: 0–5 min warmup, 5–10 stable, 10–15 early fatigue, 15–20 pronounced. Impedance stabilizes ~10–15 min.

P6 — Inter-Subject Variability: 3–5× amplitude variation. Within-subject trends far more reliable.

P7 — Amplitude Caution (Vigotsky 2018): sEMG amplitude cannot infer absolute force, compare between muscles, or determine specific activation in isolation. Valid for relative changes within the same muscle/session/electrode setup.

10. Reference Sources & Bibliography

10.1 Primary Textbooks

Merletti, R. & Parker, P. (2004). Electromyography: Physiology, Engineering, and Non-Invasive Applications. IEEE Press / Wiley.

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De Luca, C.J. (2006). Electromyography. Enc. Medical Devices and Instrumentation. Wiley.

10.2 Standards & Guidelines

SENIAM — seniam.org — Electrode placement and recording standards.

ISEK — Standards for Reporting EMG Data (Merletti). Frequency bands, sampling, definitions, smoothing.

Tankisi et al. (2020). Standards of Instrumentation of EMG. Clinical Neurophysiology.

10.3 Key Research Papers

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Campbell et al. (2022). Surface EMG Signal Quality Analysis. IEEE Reviews in Biomedical Engineering.

De Luca, C.J. (1997). Surface EMG in biomechanics. J. Applied Biomechanics, 13(2), 135–163.

Chowdhury et al. (2013). Surface EMG Signal Processing and Classification. Sensors, 13(9), 12431–12466.

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10.4 LLM + Physiological Signal Papers

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Ghobadi et al. (2025). LLM-Powered Agent for Physiological Data. arXiv:2502.12836.